

CO4 – AT1: SCENARIO-BASED ASSESSMENT

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COURSE NAME: Fundamentals Of Data Science

Question 11 – Hypothesis Testing Using Confidence Interval

Scenario: A university claims that at least 80% of its students are satisfied with online classes. A survey produces a 95% confidence interval of (74%, 79%) for the satisfaction proportion. What conclusion can be made about the university's claim.

Given	95% confidence interval = (74%, 79%)
Claim	At least 80% of students are satisfied
Confidence Level	95%
Parameter	Population proportion of satisfied students

Step 1 – State the hypotheses: Let p be the true proportion of students satisfied with online classes. $H_0: p \geq 0.80$. $H_1: p < 0.80$.

Step 2 – Compare the claim with the confidence interval: The entire 95% confidence interval, from 74% to 79%, lies below 80%. The claimed value of 80% is not contained in the interval.

Step 3 – Decision: Because 80% is outside the 95% confidence interval and the interval is entirely below 80%, there is evidence against the university's claim.

Step 4 – Interpretation: The data indicate that the true satisfaction proportion is likely below 80% at the 95% confidence level.

Conclusion: The university's claim that at least 80% of students are satisfied with online classes is not supported by the sample evidence. The claim should be rejected at the 5% significance level.

Question 12 – Hypothesis Testing Using Confidence Interval

Scenario: A mobile application developer claims that users spend an average of 30 minutes per day on the application. A sample produces a 99% confidence interval of (31.5, 35.2) minutes. Using the confidence interval, determine whether the claim is supported.

Given	99% confidence interval = (31.5, 35.2) minutes
Claim	Population mean usage time = 30 minutes/day
Confidence Level	99%
Parameter	Population mean daily usage time

Step 1 – State the hypotheses: Let μ be the true mean daily usage time. $H_0: \mu = 30$ minutes. $H_1: \mu \neq 30$ minutes.

Step 2 – Compare the claimed value with the interval: The 99% confidence interval ranges from 31.5 to 35.2 minutes. The claimed value of 30 minutes is outside this interval.

Step 3 – Decision: Since 30 minutes is not contained in the 99% confidence interval, the sample provides strong evidence that the true mean is different from 30 minutes.

Step 4 – Interpretation: The estimated mean daily usage is higher than the claimed 30 minutes, because the entire interval lies above 30 minutes.

Conclusion: The developer's claim that users spend an average of 30 minutes per day is not supported by the sample evidence. The data suggest that the true average usage is greater than 30 minutes per day.

Question 13 – Hypothesis Testing Using p-value

Scenario: A credit-card company claims that its fraud detection system identifies 95% of fraudulent transactions. After testing the system on a sample, the calculated p-value is 0.02. At a 5% significance level, what conclusion should the data scientist make.

Claim	Fraud detection system identifies 95% of fraudulent transactions
p-value	0.02
Significance Level (α)	0.05
Decision Rule	Reject H_0 if p-value $< \alpha$

Step 1 – State the hypotheses: Let p be the true proportion of fraudulent transactions correctly identified. $H_0: p = 0.95$. $H_1: p \neq 0.95$.

Step 2 – Compare p-value with α : p-value = 0.02 and $\alpha = 0.05$. Since $0.02 < 0.05$, the p-value is smaller than the significance level.

Step 3 – Decision: Reject the null hypothesis H_0 .

Step 4 – Interpretation: There is statistically significant evidence that the system's true fraud-detection rate differs from the claimed 95% rate.

Conclusion: At the 5% significance level, the data scientist should reject the company's 95% claim. The sample provides statistically significant evidence that the actual fraud-detection proportion is different from 95%.

Question 14 – Hypothesis Testing Using p-value

Scenario: An e-commerce company introduces a new recommendation algorithm. The existing algorithm has a conversion rate of 12%. After implementation, the observed conversion rate is 15%. The

hypothesis test produces a p-value of 0.03. At $\alpha = 0.05$, determine whether the new algorithm significantly improves conversion.

Existing Conversion Rate	12%
Observed Conversion Rate	15%
p-value	0.03
Significance Level (α)	0.05
Decision Rule	Reject H_0 if p-value $< \alpha$

Step 1 – State the hypotheses: Let p be the true conversion rate with the new algorithm. $H_0: p = 0.12$ (no improvement). $H_1: p > 0.12$ (the new algorithm improves conversion).

Step 2 – Compare p-value with α : p-value = 0.03 and $\alpha = 0.05$. Since $0.03 < 0.05$, the result is statistically significant.

Step 3 – Decision: Reject the null hypothesis H_0 .

Step 4 – Interpretation: There is sufficient statistical evidence that the new recommendation algorithm improves the conversion rate compared with the existing 12% rate.

Conclusion: Yes. The new recommendation algorithm significantly improves conversion at the 5% significance level. The observed conversion rate increased from 12% to 15%, and the hypothesis test indicates that this improvement is statistically significant.

Question 15 – Hypothesis Testing Using p-value

Scenario: A hospital introduces an AI-based patient scheduling system. The hospital wants to determine whether it reduces average waiting time. A hypothesis test produces a p-value of 0.08. Using a significance level of 5%, should the hospital conclude that the new system reduces waiting time?

Objective	Determine whether the AI-based scheduling system reduces average waiting time
p-value	0.08
Significance Level (α)	0.05
Decision Rule	Reject H_0 if p-value $< \alpha$

Step 1 – State the hypotheses: Let μ_n be the true mean waiting time after implementation and μ_0 the corresponding baseline mean. H_0 : the new system does not reduce average waiting time. H_1 : the new system reduces average waiting time.

Step 2 – Compare p-value with α : p-value = 0.08 and $\alpha = 0.05$. Since $0.08 > 0.05$, the p-value is greater than the significance level.

Step 3 – Decision: Fail to reject the null hypothesis H_0 .

Step 4 – Interpretation: The sample does not provide sufficient statistical evidence, at the 5% significance level, to conclude that the AI-based scheduling system reduces average waiting time.

Conclusion: No. The hospital should not conclude that the new system reduces waiting time at the 5% significance level. Although the system may show some observed effect, the evidence is not statistically significant because the p-value (0.08) is greater than 0.05.

Assessment Summary

Question	Method	Decision	Final Interpretation
11	95% CI	Reject claim	Satisfaction proportion is likely below 80%.
12	99% CI	Reject claim	Mean usage is likely above 30 minutes/day.
13	p-value	Reject H_0	Detection rate differs significantly from 95%.
14	p-value	Reject H_0	New algorithm significantly improves conversion.
15	p-value	Fail to reject H_0	Insufficient evidence that waiting time is reduced.

Prepared for CO4–AT1 Assessment | Questions 11–15